distributed randomly in the external Universe, increasing its entropy by more than the hole's entropy loss. Again the second law is obeyed.

How long does it take for a black hole to evaporate and disappear? The answer depends on the hole's mass. The larger the hole, the lower its temperature, and thus the more weakly it emits particles and the more slowly it evaporates. The total lifetime, as worked out by Don Page in 1975 when he was jointly my student and Hawking's, is 1.2×10^{67} years if the hole's mass is twice that of the Sun. The lifetime is proportional to the cube of the hole's mass, so a 20-solar-mass hole has a life of 1.2×10^{70} years. These lifetimes are so enormous compared to the present age of the Universe, about 1×10^{10} years, that the evaporation is totally irrelevant for astrophysics. Nevertheless, the evaporation has been very important for our understanding of the marriage between general relativity and quantum mechanics; the struggle to understand the evaporation taught us the laws of quantum fields in curved spacetime.

Holes far less massive than *2* Suns, if they could exist, would evaporate far more rapidly than 10^{67} years. Such small holes cannot be formed in the Universe today because degeneracy pressures and nuclear pressures prevent small masses from imploding, even if one squeezes them with all the force the present-day Universe can muster (Chapters 4 and 5). However, such holes might have formed in the big bang, where matter experienced densities and pressures and gravitational squeezes that were enormously higher than in any modern-day star.

Detailed calculations by Hawking, Zel'dovich, Novikov, and others have shown that tiny lumps in the matter emerging from the big bang could have produced tiny black holes, if the lumps' matter had a rather soft equation of state (that is, had only small increases of pressure when squeezed). Powerful squeezing by other, adjacent matter in the very early Universe, like the squeezing of carbon in the jaws of a powerful anvil to form diamond, could have made the tiny lumps implode to produce tiny holes.

A promising way to search for such tiny *primordial black holes* is by searching for the particles they produce when they evaporate. Black holes weighing less than about 500 billion kilograms (5×10^{14} grams, the weight